Prompt 4. Finance and Marketing: Find What's Wrong, Then Trust Nothing

invoices.csv (603 rows, Jul 2025 to Jun 2026) and marketing_campaigns.csv (750 rows, Jan to May 2026).

A. Invoice audit, acting as AP auditor

I asked the AI to flag duplicates, outliers, and inconsistencies, then checked every flag against the raw data myself, and ran a few checks the AI hadn't proposed (date logic, vendor name consistency, payment terms by category) to see whether anything real was missed.

Confirmed anomalies

Flag	Evidence	Verdict
Duplicate payment	INV-50101 and INV-50101D: identical vendor (LawFirm Horvat), amount (EUR 3,510.20), issue date, due date and description. Both marked paid, both with the same payment_date.	Real. The "D" suffix on the second ID, combined with every other field matching exactly, points to an actual double payment already made, not just a duplication risk to watch for.
Billing outlier	Notion invoices INV-59001 (EUR 3,934.48, issued 2025-08-29) and INV-59002 (EUR 2,880.00, issued 2025-11-05), both labelled "Monthly subscription". Every other Notion invoice (56 more, spanning the full year) falls between EUR 20.02 and EUR 399.48, median EUR 220.75.	Real. These two are 10 to 30 times the normal Notion charge for the identical description, and use a different invoice-ID numbering pattern (59xxx instead of 50xxx), which is itself a sign they entered the ledger a different way than the rest.
Status mislabeling	All 6 invoices currently marked "pending" (HEP Elektra, Bean Supplier, Google Ads, Uber x2, Deloitte Advisory) have due dates 40 to 52 days before today (2026-07-06), with no payment_date recorded. That is the same objective condition as the 149 invoices already marked "overdue".	Real, and the kind of thing a status-column read would miss. Adds EUR 5,308.01 to true cash at risk: EUR 168,308.29 total, not the EUR 163,000.28 the status field alone shows.

What a naive pass would invent or miss

Would invent: treating the 156 invoices that were paid late in the past (EUR 179,234.65, already collected) as if they were current risk. They are resolved cash flow history, not exposure, and shouldn't be folded into the EUR 168,308.29 figure above just because "late" sounds alarming.
Would miss: the pending-vs-overdue mislabeling above, since it requires comparing due_date against today rather than trusting the status field as given.

Checks that came back clean

For completeness: no negative or zero invoice amounts; no vendor name variants (13 vendors, single spelling each); no invoice where issue_date falls after due_date; no invoice paid before it was issued; payment terms are a consistent ~30 days across all six categories with no category-specific irregularity; and the three largest invoices overall (up to EUR 5,980.38) are ordinary Professional Services consulting fees, not billing errors, since that category runs highest by nature. Nothing else in the file looked anomalous beyond the three flags above.

B. Marketing ROAS: real star performer or attribution illusion

I asked the AI to compute spend, revenue and ROAS by channel and write a reallocation recommendation. It highlighted Email's 88.8x ROAS as the standout and proposed shifting budget toward it. I fact-checked that number before accepting the recommendation.

Channel	Spend	Revenue	ROAS	Conversions	CPA
Email	EUR 18,718	EUR 1,662,631	88.8x	54,412	EUR 0.34
Google Ads	EUR 207,388	EUR 298,212	1.44x	7,773	EUR 26.68
Meta	EUR 69,108	EUR 68,340	0.99x	1,972	EUR 35.04
LinkedIn	EUR 276,480	EUR 151,157	0.55x	1,254	EUR 220.48
Organic	EUR 0	EUR 48,998	undefined	1,322	EUR 0.00

Why the number is misleading

Three things rule out "Email is genuinely 60 times more efficient than Google Ads":

1. **Cost scope.** Email's implied cost per click is EUR 0.021, consistent with a per-send platform fee only. There is no line for list acquisition, content production, or platform subscription, costs that Google Ads and Meta's spend figures do include by nature of being pay-per-click auctions.
2. **Attribution.** Email converts 54,412 times, roughly 7 times more than Google Ads on a fraction of the budget. Average order value is EUR 30.56, in the same range as every other channel (EUR 30 to 40), so this isn't unusually large purchases, it's unusually many, which is what you'd expect if a last-touch model is crediting Email for purchases a customer was already going to make after being reached first by paid or organic channels.
3. **Scalability.** Email spend and revenue are flat month over month (EUR 3,100 to 4,300 per month, Jan to May 2026), consistent with a fixed subscriber list being emailed on a routine cadence, not a channel where more budget buys more reach the way an ad auction does. Organic shows the same distortion in its purest form: EUR 0 recorded spend against EUR 48,998 in revenue is not "infinite ROAS", it's an unpriced cost.

Corrected commentary and reallocation

Do not shift budget "into Email" as the 88.8x figure would naively suggest: it is not a lever that scales with more money, and crediting it with more of the story overstates its actual incremental contribution. The channels that are genuinely comparable on a like-for-like, pay-to-play basis are Google Ads, Meta and LinkedIn, and among those:

1. **Cut LinkedIn.** It is the clear underperformer (ROAS 0.55, CPA EUR 220.48, almost 8 times Google's CPA) despite having the second-largest budget of all five channels (EUR 276,480, more than Google Ads).
2. **Shift the freed budget to Google Ads,** the best-performing scalable, paid channel (ROAS 1.44, CPA EUR 26.68), and monitor for diminishing returns as spend increases.
3. **Invest part of the LinkedIn cut into list growth** (referral programs, on-site sign-up prompts) so more customers flow into the Email channel over time, rather than assuming Email itself can simply absorb a bigger budget.
4. **Fix attribution before the next review.** Move from last-touch to a multi-touch or data-driven model so Email and Organic are credited only for the incremental revenue they actually generate, which will bring their reported ROAS down to a comparable, trustworthy number.